

Various methods of Central Tendency.

- arithmetic mean
- mode
- harmonic mean
- geometric mean

Q. As a part of psychological assessment aimed at understanding student learning potential a counsellor recorded the intelligent quotient of 10 boys from a particular class. The purpose of the assessment was to analyze the cognitive ability levels and tell her the features and the IQ scores recorded were as follows:

17, 120, 110, 101, 88, 83, 75, 98, 107, 100

Determine the average.

$$\text{avg} = \frac{899}{10} = 89.9$$

17.3.2026

Q. A following is the frequency distribution of number of telephone calls received in 245 successive one min interval at an exchange.

no. of calls	freq
0	14
1	21
2	25
3	43
4	51
5	40
6	39
7	12

Obtain the mean number of calls per min.

Q. 10

x	f	fx
0	14	0
1	21	21
2	25	50
3	43	129
4	51	204
5	40	200
6	39	234
7	12	84

$$N = 245$$

$$\bar{X} = \frac{\sum fx}{N} = \frac{922}{245} = 3.76$$

$$\sum fx = 922$$

Step deviation method

$$\bar{X} = A + h \frac{\sum fd}{N}$$

$$d = \frac{x - A}{h}$$

A - arbitrary value (usually mid value)

Steps to compute mean

- 1) compute $d = \frac{x - A}{h}$; A $\rightarrow$ any arbitrary value
h $\rightarrow$ magnitude of class
- 2) multiply ~~mean~~ corresponding frequency f to get fd
- 3) Find the sum of product obtained in step 2 to get summation of fd
- 4) Divide sum obtained in step 3 by N where N is total frequency
- 5) Multiply value obtained in step 4 by h
- 6) Add A to value obtained in step 5

Problems

- 1) In an effort to evaluate in overall academic performance of students in math exam, a school compiled the scores and group them into the following. The school administration is interested in determining the average performance of the class. For this purpose arithmetic mean is considered an appropriate measure of central tendency as it provides straight forward representation of average score obtained. Calculate the arithmetic mean

$A = 35$

x marks	f no. of students	X	$d = \frac{x-A}{h}$	fd
0-10	6	$\frac{0+10}{2} = 5$	$\frac{5-35}{10} = -3$	-15
10-20	5	$\frac{10+20}{2} = 15$	-2	-10
20-30	8	25	-1	-8
30-40	15	<u>35</u>	0	0
40-50	7	45	1	7
50-60	6	55	2	12
60-70	3	65	3	9

$$\Sigma f = 50$$

$$\Sigma fd = -8$$

$$N = \Sigma f = 50$$

$$\bar{X} = A + h \frac{\Sigma fd}{N} = 35 + \frac{10 \times -8}{50} = 33.4$$

2. The numbers 3.2, 5.8, 7.9, 4.5 have the frequencies x , $x+2$, $x-3$, $x+6$ respectively, if the arithmetic mean is 4.876. find the value of x .

(x) no. x	f	fx
3.2	x	$3.2x$
5.8	$x+2$	$5.8x + 11.6$
7.9	$x-3$	$7.9x - 23.7$
4.5	$x+6$	$4.5x + 27$

$$\bar{X} = \frac{\Sigma fx}{N}$$

$$N = \Sigma f$$

$$4.876 = \frac{21.4x + 14.9}{4x + 5}$$

$$19.504x + 24.38 = 21.4x + 14.9$$

$$9.48 = 1.9x$$

$$x = 5$$

3. From the following data of income distribution calculate the arithmetic mean, Given that

- i) Total income of the persons in the highest group £435
 ii) None of them ^{is} earning less than £20

Income	No of persons
Below 30	16
Below 40	36
50	61
60	76
70	87
80	95
80 & over	5

class interval	(midval) X	no. of (f) persons	$f \times X$
20 - 30	25	16	400
30 - 40	35	20	700
40 - 50	45	25	1125
50 - 60	55	15	825
60 - 70	65	11	715
70 - 80	75	8	600
80 & over	-	5	435

$$\Sigma f = 100 \quad \Sigma fX = 4800$$

$$\bar{X} = \frac{4800}{100} = 48$$

Weighted AM

$$\bar{X}_w = \frac{\sum w_i x_i}{\sum w_i} = \frac{\sum w(fx)}{\sum w}$$

1. A candidate obtained the following percentages of marks in an exam English 60, Hindi 75, Math 63, Physics 59 and Chemistry 55. Find the candidate's weighted ~~arithmetic~~ Arithmetic mean if weights are 1, 2, 1, 3 and 3 respectively are allotted for the subjects.

Sub	(x) marks	(w) weights	wx
English	60	1	60
Hindi	75	2	150
Math	63	1	63
Physics	59	3	177
Chemistry	55	3	165
$\sum w = 10$			$\sum wx = 615$

$$\begin{aligned}\bar{X}_w &= \frac{\sum wx}{\sum w} \\ &= \frac{615}{10}\end{aligned}$$

$$= 61.5$$

Mathematical Properties of AM

1. The algebraic sum of the deviations of the given set of observations from their arithmetic mean is zero

$$\sum f(x - \bar{x}) = 0$$

2. If n_1 and n_2 are the sizes. $\bar{x}_1$ and $\bar{x}_2$ are respective means of two groups then mean $\bar{x}$ of the combined group of size $(n_1 + n_2)$ is given by

$$\bar{x} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2}$$

3. The sum of the squares of the deviation of the given set of observations is minimum when taking from A.M.

$$S = \sum f (X - A)^2$$

4. ~~The mean of marks~~

Q. The mean of marks in statistics of 100 students in a class was 72. The mean of marks of ~~boys~~ boys is 75, while their number was 70. Find out the mean marks of girls in a class.

$$n_1 = 70 \quad n_2 = 30 \quad \bar{x}_1 = 75 \quad \bar{x}_2 = ? \quad \bar{X} = 72$$

$$\bar{X} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2}$$

$$72 = \frac{70 \times 75 + 30 \bar{x}_2}{70 + 30}$$

$$30 \bar{x}_2 = 75(70 + 30) - (70 \times 72)$$

$$7200 = 5050 + 30 \bar{x}_2$$

$$30 \bar{x}_2 = 2150$$

$$\boxed{\bar{x}_2 = 71.65}$$

Median

The value of variable which divides the group into two equal parts comprising of all the values greater and other all the values less than the median.

$$\text{median} = l + \frac{h}{f} \left(\frac{N}{2} - c \right)$$

where l is the lower limit of the median class
 f is the frequency of median class
 h is the magnitude or width of the median class
 c is the cumulative frequency of the class preceding median class

1. The following distribution shows the age distribution of persons in a particular regions

Age (in years)	below 10	below 20	below 30	below 40	below 50	below 60	below 70	70 & over
No. of persons	2	5	9	12	14	15	15	16

Age	No. of persons	Cf
0-10	2	2
10-20	3	5
20-30	4	9
30-40	3	12
40-50	2	14
50-60	1	15
60-70	0	15
70 & over	1	16

$$\frac{N}{2} = \frac{16}{2} = 8$$

Ex. Sol.
 [cf near to $N/2$ select it]

$$\text{median} = l + \frac{h}{f} \left(\frac{N}{2} - c \right)$$

$$= 20 + \frac{10}{4} \left[\frac{16}{2} - 5 \right]$$

$$= 27.5$$

Note: The value which occurs

Mode: The value that occurs most frequently in a set of observations

$$\text{mode} = \frac{l + \frac{n(f_1 - f_0)}{(f_1 - f_0) - (f_2 - f_1)}}{2}$$

$$\text{mode} = l + \frac{n(f_1 - f_0)}{2f_1 - f_0 - f_2}$$

where $l \rightarrow$ lower limit of class

$f_1 \rightarrow$ frequency of class which is max frequency

$f_0 \rightarrow$ frequency of class preceding f_1 class

$f_2 \rightarrow$ frequency of class succeeding f_1 class

$n \rightarrow$ magnitude of the class

Q. 3. 26

Empirical Solution

$$\text{mode} = \text{mean} - 3(\text{mean} - \text{median})$$

$$\text{mean} - \text{mode} = 3(\text{mean} - \text{median})$$

$$\text{mean} - \text{median} = \frac{1}{3}(\text{mean} - \text{mode})$$

inclusive class

10-20
20-30

exclusive class

10-20
21-30

Find the mean median & mode from the data
 $A = 110$ (given)

weight	93-97	98-102	103-107	108-112	113-117	118-122	123-127	128-132
no. of students	3	5	12	17	14	6	3	1

as exclusive we need to convert to inclusive to find mean mode & median

weight	X	$d = \frac{X-A}{h}$	f	fd	X	cf
92.5 - 97.5	95	-3	3	-9	3	
97.5 - 102.5	100	-2	5	-10	8	
102.5 - 107.5	105	-1	12	-12	20	
107.5 - 112.5	110	0	17	0	37	
112.5 - 117.5	115	1	14	14	51	
117.5 - 122.5	120	2	6	12	57	
122.5 - 127.5	125	3	3	9	60	
127.5 - 132.5	130	4	1	4	61	

$$\Sigma f = 61$$

$$\Sigma fd = 8$$

$$q: \bar{X} = A + h \frac{\Sigma fd}{N}$$

$$= 110 + 5 \left(\frac{8}{61} \right)$$

$$= 110.656$$

$$\text{median} = l + \frac{h}{f} \left(\frac{N}{2} - c \right)$$

$$\frac{N}{2} = \frac{61}{2} = 30.5$$

$$= 107.5 + \frac{5}{17} \left(\frac{61}{2} - 20 \right)$$

$$= 110.588$$

$$f_1 = 17, f_0 = 12, f_2 = 14$$

$$\text{mode} = l + \frac{h(f_1 - f_0)}{2f_1 - f_0 - f_2}$$

$$= 107.5 + \frac{5(17 - 12)}{2(17) - 12 - 14}$$

$$= 110.625$$

$$\text{mean} - \text{mode} = 3(\text{mean} - \text{median})$$

$$110.656 - \text{mode} = 3(110.656 - 110.588)$$

$$\text{mode} = \text{mean} - 3(\text{mean} - \text{median})$$

$$= 110.656 - 3(110.656 - 110.588)$$

$$= 110.452$$

2. Calculate the median of the following distribution

annual sales	frequency
less than 10	4
less than 20	20
less than 30	35
less than 40	55
less than 50	62
less than 60	67

Annual Sales	X	f	cf	$d = \frac{1-f}{h}$	fd
0-10	5	4	4	-20	-120
10-20	15	20-4=16	20	-20	-320
20-30	25	15	35	-10	-150
30-40	35	20	55	0	0
40-50	45	7	62	10	70
50-60	55	5	67	20	100

$\Sigma fd = 42$

$$\text{median} = l + \frac{h}{f} \left(\frac{n}{2} - c \right)$$

$$\frac{n}{2} = \frac{67}{2} = 33.5$$

$$= 20 + \frac{10}{15} \left(\frac{67}{2} - 20 \right)$$

$$= 29$$

$$\text{mean} = A + h \frac{\Sigma fd}{N}$$

$$\text{mode} = \text{mean} - 3(\text{mean} - \text{med})$$

$$= 28.75 - 3(28.75 - 29)$$

$$= 29.56$$

$$l_1 = 20 \quad f_0 = 15 \quad f_2 = 7$$

$$\text{mode} = l + \frac{h(f_1 - f_0)}{2f_1 - f_0 - f_2}$$

$$= 20 + \frac{10(20 - 15)}{40 - 15 - 7} = 32.7$$

3. A secondary school maths teacher instructor recorded the performance of the students in recent assessment and group the marks obtained into the following class interval the teacher aims to conduct a statistical analysis to better understand the overall performance of the class while the teacher decides to compute geometric mean of the marks to account for skewed nature of the data so determine the GM.

Geometric Mean

$$\log(GM) = \frac{1}{n} \sum \log x$$

$$GM = \text{antilog} \left[\frac{1}{n} \sum \log x \right]$$

$$\& \text{antilog} \left[\frac{1}{n} \sum \log x \right]$$

Marks	No. of Stu	X	log X	f log X
0-10	5	5	0.6989	3.4945
10-20	7	15	1.1761	8.232
20-30	15	25	1.3979	20.9685
30-40	25	35	1.544	38.6
40-50	8	45	1.6532	13.2256
$\Sigma f =$			$\Sigma \log x = 6.47$	$\Sigma f \log x = 84.5206$

$$GM = \text{antilog} \left[\frac{1}{n} \sum f \log x \right]$$

$$= \text{antilog} \left[\frac{1}{60} \times 84.5206 \right]$$

$$= 25.625$$

25.3.26

Harmonic Mean

$$H = \frac{N}{\sum (f/x)}$$

1. A Health research organisation conducted a survey including 31 individuals to analyse general weight trends in a local community. The data collected is intended to inform future wellness program by identifying typical weight programs amongst the residents. As a data analyst you are provided with weights of 31 individuals participated in the survey. Calculate GM & HM

weights	130	135	140	145	146	148	149	150	157
No. of people	3	4	6	6	3	5	2	1	1

X	f	log X	f log X	f/x
130	3	2.114	6.342	0.023
135	4	2.130	8.52	0.030
140	6	2.146	12.876	0.043
145	6	2.161	12.966	0.041
146	3	2.164	6.492	0.021
148	5	2.170	10.85	0.034
149	2	2.173	4.346	0.013
150	1	2.176	2.176	0.006
157	1	2.196	2.196	0.006
$\Sigma f = 31$			$\Sigma f \log x = 66.764$	$\Sigma f/x = 0.217$

$$\Sigma f \log x = 66.764 \quad \Sigma f/x = 0.217$$

$$\begin{aligned}
 GM &= \text{antilog} \left[\frac{1}{N} \Sigma f \log x \right] \\
 &= \text{antilog} [2.1537] \\
 &= 142.4623
 \end{aligned}$$

$$\begin{aligned}
 HM &= \frac{N}{\Sigma (f/x)} = \frac{31}{0.217} \\
 &= 142.857
 \end{aligned}$$

Measure of Dispersion

1. Range

$$\begin{aligned}
 \text{co. eff of Range} &= \frac{X_{\max} - X_{\min}}{X_{\max} + X_{\min}}
 \end{aligned}$$

$$\text{Range} = X_{\max} - X_{\min}$$

2. Quartile deviation
3. Mean deviation
4. Standard deviation

2. Quartile deviation

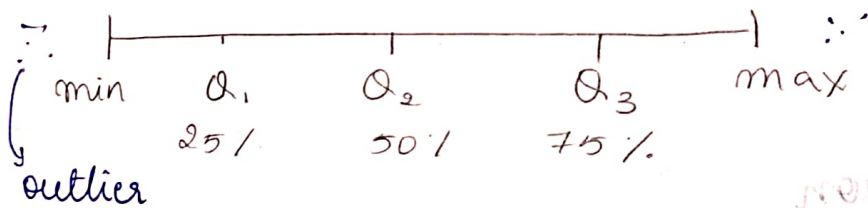
Inter-Quartile range = $Q_3 - Q_1$

Quartile deviation = $\frac{Q_3 - Q_1}{2}$

co eff of QD = $\frac{Q_3 - Q_1}{Q_3 + Q_1}$

1. Range: It is the difference between the upper limit of highest class and lower limit of smallest class.

5 no. Summary



Absolute measure: It is expressed in terms of original units of series whereas relative measures is obtained as ratios or percentages.

1. Calculate the range and coefficient of range of a persons monthly earning for a year

month	1	2	3	4	5	6	7	8	9	10	11	12
monthly earning	139	150	151	154	157	158	160	161	162	162	173	175

Range = $X_{max} - X_{min}$
 $= 17500 - 13900$
 $= 3600$

coeff of Range = $\frac{X_{max} - X_{min}}{X_{max} + X_{min}}$
 $= 0.1146$

§ Quartile deviation : Interquartile ~~deviation~~ range
 Is a measure of dispersion based on upper
 quartile and lower quartile ($Q_3 - Q_1$)

ungrouped data

$$Q_1 = \frac{n+1}{4}$$

grouped data

$$Q_1 = l + \frac{h}{f} \left[\frac{N}{4} - c \right] \quad Q_3 = l + \frac{h}{f} \left[\frac{3N}{4} - c \right]$$

1. Find

- i) Inter quartile
- ii) quartile deviation
- iii) coeff of quartile deviation

class Interval	f	cf
0-15	8	8
15-30	26	34
30-45	30	64
45-50	45	109
50-75	20	129
75-90	17	146
90-105	4	150

$$l = 30 \quad h = 15 \quad f = 30 \quad c = 34$$

$$Q_1 = l + \frac{h}{f} \left[\frac{N}{4} - c \right]$$

$$= 30 + \frac{15}{30} \left[\frac{150}{4} - 34 \right]$$

$$= 31.75$$

$$Q_3 = l + \frac{h}{f} \left[\frac{3N}{4} - c \right]$$

$$= 30 + \frac{15}{30} \left[\frac{3 \times 150}{4} - 34 \right]$$

$$= 69.25$$

$$\Sigma f = 150$$

$$\Sigma f = 150$$

$$N/4 = \frac{150}{4} = 37.5$$

$$N/4 = \frac{150}{4} = 37.5$$

$$\frac{3N}{4} = 112.5$$

$$l = 60 \quad h = 15$$

$$l = 60 \quad h = 15 \quad f = 20 \quad c = 109$$

$$\begin{aligned} Q_3 &= l + \frac{h}{f} \left[\frac{3N}{H} - c \right] \\ &= 60 + \frac{15}{20} \left[\frac{3 \times 150}{H} - 109 \right] \\ &= 62.625 \end{aligned}$$

i) Inter quartile range

$$Q_3 - Q_1 = 30.875$$

ii) quartile deviation

$$\frac{Q_3 - Q_1}{2} = 15.437$$

iii) coeff of quartile deviation $\frac{Q_3 - Q_1}{Q_3 + Q_1} = 0.3271$

26-3-26

2. From the following data find
Quartile deviation

c I	f	c f
0-10	3	3
10-20	5	8
20-30	7	15
30-40	9	24
40-50	4	28

$$N = 28 \quad \frac{N}{2} = \frac{28}{2} = 14$$

$$\frac{3N}{4} = \frac{3 \times 28}{4} = 21$$

$$l = 10 \quad h = 10 \quad f = 5 \quad c = 3$$

$$\begin{aligned} Q_1 &= l + \frac{h}{f} \left[\frac{N}{4} - c \right] \\ &= 10 + \frac{10}{5} \left[\frac{28}{4} - 3 \right] \\ &= 18 \end{aligned}$$

$$l = 30 \quad h = 10 \quad f = 9 \quad c = 15$$

$$\begin{aligned} Q_3 &= l + \frac{h}{f} \left[\frac{3N}{4} - c \right] \\ &= 30 + \frac{10}{9} \left[\frac{3 \times 28}{4} - 15 \right] \\ &= 36.67 \end{aligned}$$

$$\text{quartile deviation} = \frac{Q_3 - Q_1}{2} = \frac{36.67 - 18}{2} = 9.335$$

3. Evaluate an appropriate measure of dispersion for the following

Income	less than 50	50-70	70-90	90-110	110-130	130-150	above 150
No. of persons	54	100	140	300	230	125	51

Comment the following:

If coefficient of quartile deviation is 0.6 and quartile deviation is 15 then Q_1 is 10 Q_3 is 40

CI	f	cf
less than 50	54	54
50-70	100	154
70-90	140	294
90-110	300	594
110-130	230	824
130-150	125	949
above 150	51	1000

$$\frac{N}{h} = \frac{1000}{h} = 250$$

$$\frac{3N}{h} = \frac{3 \times 1000}{h} = 750$$

$$d = 110 \quad h = 20$$

$$f = 230 \quad cf = 824$$

$$Q_1 = d + \frac{h}{f} \left[\frac{N}{4} - c \right]$$

$$d = 70 \quad f = 140 \quad h = 20 \quad c = 154$$

$$Q_1 = 70 + \frac{20}{140} \left[\frac{1000}{4} - 154 \right]$$

$$Q_1 = 76.857$$

$$Q_3 = d + \frac{h}{f} \left[\frac{3N}{4} - c \right]$$

$$= 110 + \frac{20}{230} \left[\frac{3 \times 1000}{4} - 824 \right]$$

$$= 123.565$$

$$= 123.565$$

$$QD = \frac{Q_3 - Q_1}{2}$$

$$15 = \frac{Q_3 - Q_1}{2}$$

$$Q_3 - Q_1 = 30$$

$$\text{coeff} = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

$$0.6 = \frac{30}{Q_3 + Q_1}$$

$$Q_3 + Q_1 = \frac{30}{0.6} = 50$$

$$Q_3 - Q_1 = 30$$

$$Q_3 + Q_1 = 50$$

$$2Q_3 = 80$$

$$Q_3 = 40$$

$$Q_3 - Q_1 = 30$$

$$40 - Q_1 = 30$$

$$Q_1 = 10$$

But the values obtained using data is $Q_1 = 83.71$ & $Q_3 = 113.565$. Therefore the given statement is false.

Mean Deviation

$$MD \text{ (about mean)} = \frac{1}{N} \sum f |x - m|$$

$$MD \text{ (about median)} = \frac{1}{N} \sum f |x - md|$$

$$MD \text{ (about mode)} = \frac{1}{N} \sum f |x - median|$$

$$\text{Coeff of MD} = \frac{\text{mean deviation}}{\text{avg about which is calculated}}$$

$$\text{Coeff of MD about Mean} = \frac{MD}{\text{mean}}$$

$$\text{Coeff of MD about median} = \frac{MD}{\text{median}}$$

1. Calculate the mean deviation about mean for the following data

CI	f	X	fx	$ x - \bar{x} $	$f x - \bar{x} $
2-4	3	3	9	2.2	6.6
4-6	4	5	20	0.2	0.8
6-8	2	7	14	1.8	3.6
8-10	1	9	9	3.8	3.8
	$\Sigma f = 10$		$\Sigma fx = 52$		$\Sigma f x - \bar{x} = 14.8$

$$\bar{x} = \frac{\Sigma fx}{\Sigma f} = \frac{52}{10} = 5.2$$

$$MD = \frac{1}{N} \sum f |x - \bar{x}| = 1.48$$

2. Calculate the mean deviation from median of the following distribution. Also find coeff of mean deviation from median

ICI	f	cf	x - median	f(x - median)	$ x - \text{median} $
50 - 100	7	7	75	125	87.5
100 - 150	18	25	125	75	135.0
150 - 200	25	50	175	25	62.5
200 - 250	31	81	225	25	77.5
250 - 300	15	96	275	75	112.5
300 - 350	4	100	325	125	50.0

$$\Sigma = 5250$$

$$\frac{N}{2} = \frac{100}{2} = 50$$

$$\text{median} = d + \frac{h}{f} \left[\frac{N}{2} - c \right]$$

$$d = 200 \quad h = 50 \quad f = 31 \quad c = 50$$

$$\text{median} = 200 + \frac{50}{31} \left[\frac{100}{2} - 50 \right]$$

$$= 200$$

$$MD = \frac{1}{N} \Sigma f|x - \text{median}|$$

$$= 52.5$$

$$\text{coeff of MD abt median} = \frac{MD}{\text{median}} = \frac{52.5}{200}$$

$$= 0.2625$$

31.3.2026

3. Find the mean deviation from mean

x	f	fx	$ x - \bar{x} $	f(x - \bar{x})	$f x - \bar{x} $
1	3	3	3.24	9.72	9.72
2	5	10	2.24	11.2	11.2
3	8	24	1.24	9.92	9.92
4	12	48	0.24	2.88	2.88
5	10	50	0.76	7.6	7.6
6	7	42	1.76	12.32	12.32
7	5	35	2.76	13.8	13.8

$$\Sigma f = 50$$

$$\Sigma fx = 212$$

$$\Sigma f|x - \bar{x}| = 67.44$$

$$\bar{x} = \frac{\Sigma fx}{N} = \frac{212}{50} = 4.24$$

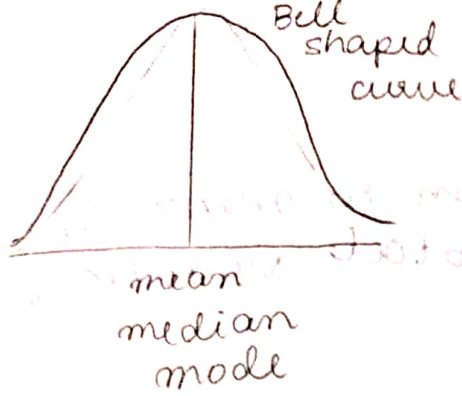
$$MD(\text{abt mean})$$

$$= \frac{1}{N} (\Sigma f|x - \bar{x}|)$$

$$= \frac{1}{50} (67.44)$$

$$= 1.3488$$

Std deviation : It is a measure of central ^{to} dependancy that defines how data value deviate from mean.

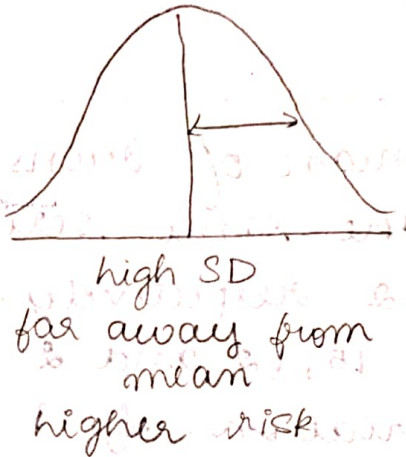
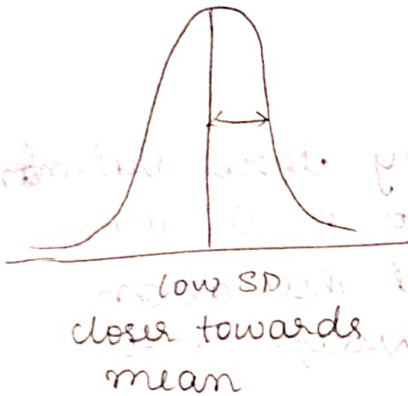


eg: 10, 20, 30, 50, 55

SD = 20 $\bar{x} = 25$ low SD

eg: 5, 6, 7, 8, 9

SD = 8 $\bar{x} = 16$ high SD



$$\sigma = \sqrt{\frac{1}{n} \sum (x - \bar{x})^2}$$

$$\text{variance } \sigma^2 = \frac{1}{n} \sum (x - \bar{x})^2$$

1. Calculate the SD for the following observation on certain variable 240.12, 240.13, 240.15, 240.12, 240.17, 240.15, 240.17, 240.16, 240.22, 240.21

$N = 10$

$$\bar{x} = \frac{\text{Sum of observation}}{\text{total no. of observation}} = \frac{2401.6}{10} = 240.16$$

240.12
240.13
240.15
240.12
240.17
240.15
240.17
240.16
240.22
240.21

X	$(x - \bar{x})$	$(x - \bar{x})^2$
240.12	0.04	1.6×10^{-3}
240.13	0.03	9×10^{-3}
240.15	0.01	1×10^{-3}
240.12	0.04	1.6×10^{-3}
240.17	0.01	1×10^{-3}
240.15	0.01	1×10^{-3}
240.17	-0.01	1×10^{-3}
240.16	0	0
240.22	-0.06	3.6×10^{-3}
240.21	-0.05	2.5×10^{-3}

$$SD = \sqrt{\frac{0.0106}{10}}$$

$$\sigma = 0.03$$

Inference: High standard deviation

$$s = 10.6 \times 10^{-3} = 0.0106$$

Coeff of SD deviation

$$\text{coeff of SD} = \frac{\sigma}{\bar{X}}$$

coeff of variation is % validation in mean & std deviation being considered as total variation in mean.

$$CV = 100 \times \frac{\sigma}{\bar{X}}$$

1. The arithmetic mean of runs scored by three batsmen A, B and C in the same series of ~~10~~ 10 innings are 50, 48 and 12 respectively. the std deviation their runs are 15, 12 and 2 respectively. Find the coeff of variation of 3.

$$\begin{aligned} X_1 &= 50 \\ X_2 &= 48 \\ X_3 &= 12 \end{aligned}$$

$$\bar{X} = \frac{X_1 + X_2 + X_3}{3}$$

$$CV_1 = 100 \times \frac{\sigma_1}{\bar{X}_1} = 100 \times \frac{15}{50} = 30$$

$$CV_2 = 100 \times \frac{\sigma_2}{\bar{X}_2} = 100 \times \frac{12}{48} = 25$$

$$CV_3 = 100 \times \frac{\sigma_3}{\bar{X}_3} = 100 \times \frac{2}{12} = 16.67$$

2. From the prices x and y of shares A and B respectively given below. Find cv.

prices of share A	55	54	52	53	56	58	52	50	51	49
Price of share B	108	107	105	105	106	107	104	103	104	101

x prices of share A	y prices of share B	$(x - \bar{x})^2$	$(y - \bar{y})^2$	$(y - \bar{y})^2$
55	108	4	184.76	9
54	107	1	158.76	4
52	105	1	112.36	0
53	105	0	172.36	0
56	106	9	134.56	1
58	107	25	158.76	4
52	104	1	92.16	7
50	103	9	73.96	4
51	104	4	92.16	1
49	101	16	43.56	16
$\Sigma(x - \bar{x})^2 = 70$			$\Sigma = 40$	

$$\bar{x} = \frac{530}{10} = 53$$

$$\bar{y} = \frac{1050}{10} = 105$$

$$\begin{aligned}\sigma^2 &= \frac{1}{n} \Sigma (x - \bar{x})^2 \\ &= \frac{1}{10} (70) \\ &= 7\end{aligned}$$

$$\begin{aligned}\sigma^2 &= \frac{1}{n} \Sigma (y - \bar{y})^2 \\ &= \frac{1}{10} (40) \\ &= 4\end{aligned}$$

$$\sigma = 2$$

$$\sigma = 2.645$$

$$\begin{aligned}CV &= 100 \times \frac{\sigma}{\bar{x}} \\ &= 100 \times \frac{2.645}{53} \\ &= 4.99\end{aligned}$$

$$\begin{aligned}CV &= 100 \times \frac{\sigma}{\bar{y}} \\ &= 100 \times \frac{2}{105} \\ &= 1.905\end{aligned}$$

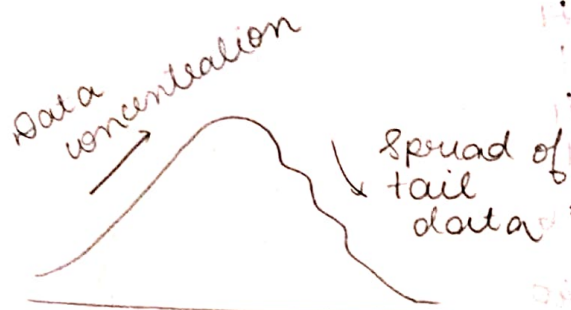
Skewness

Lack of symmetry

Normal

+ve skewness

-ve skewness



+ve skewness

> 0

mean $>$ median $>$ mode

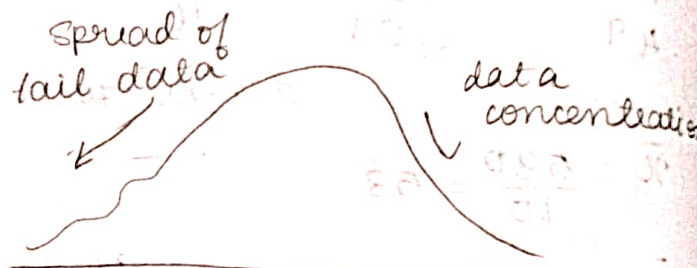
eg: income distribution



Normal Skewness

$= 0$

mean = mode = median



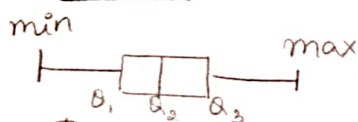
-ve skewness

< 0

mean $<$ median $<$ mode

eg: few people retire early.

Outliers



Box plot

used to find outliers.

~~xx~~ viva causes of outliers

- Data entry errors
- Measurement errors
- intentional errors
- Sampling errors
- Natural errors



left skewed



right skewed

1 Lower Bound = $Q_1 - 1.5 IQR$

Upper Bound = $Q_3 + 1.5 IQR$

Methods to detect Outliers

- IQR Method inter quartile range
- 5 number summary
- Boxplot method
- ZOR method

$$Z = \frac{X - \mu}{\sigma} \quad \text{mean}$$

Kurtosis

Tailiness of distributions

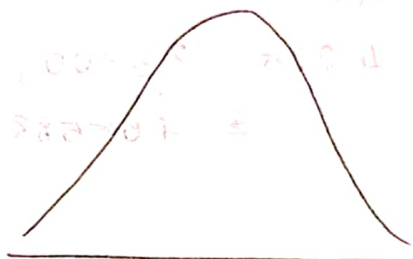


mesokurtic

$$= 3$$

eg: exam scores

heights & weights of people



leptokurtic

$$> 3$$

more outliers
heavy tail

eg: exam scores where many students scored either very high / very low



platykurtic

$$< 3$$

eg: scores by students with the same marks

Uses of Kurtosis:

- Detection of outliers

Python:

from scipy.stats import kurtosis

$$\text{skewness} = \frac{3 \times (\text{mean} - \text{mode})}{\text{std deviation}}$$

$$SD = \sqrt{\frac{\sum fx^2}{N} - \left(\frac{\sum fx}{N}\right)^2}$$

1. Compute Karl Pearson's coefficient of skewness from the following data. Assumed mean is 61.

height	58	59	60	61	62	63	64	65
no. of persons	10	18	30	42	35	28	16	8

X	f	fx	x ²	fx ²
58	10	580	3364	33640
59	18	1062	3481	62658
60	30	1800	3600	108000
61	42	2562	3721	156282
62	35	2170	3844	134540
63	28	1764	3969	111132
64	16	1024	4096	65536
65	8	520	4225	33800
Σf	187	$\Sigma = 11482$		$\Sigma = 705588$

$$\bar{X} = \frac{\Sigma fx}{N}$$

$$= \frac{11482}{187}$$

$$= 61.401$$

$$SD = \sqrt{\frac{\Sigma fx^2}{N} - \left(\frac{\Sigma fx}{N}\right)^2}$$

$$= \sqrt{\frac{705588}{187} - \left(\frac{11482}{187}\right)^2}$$

$$= 1.7625$$

INCOMPLETE

Incomplete

Incomplete

2. Consider the following distribution.

	Distribution A	Distribution B
mean	100	90
median	90	80
SD	10	10

a) Is distribution A has same degree of variation as distribution B

b) State true or false, does both distribution have the same degree of skewness.

$$CV_A = 100 \times \frac{\sigma}{\bar{X}}$$

$$= 100 \times \frac{10}{100}$$

$$= 10$$

$$CV_B = 100 \times \frac{\sigma}{\bar{X}}$$

$$= 100 \times \frac{10}{90}$$

$$= 11.111$$

Not same degree of variation.

~~skewness~~ $A = 3$

$$\text{mode} = \text{mean} - 3(\text{mean} - \text{median})$$

$$= 100 - 3(100 - 90)$$

$$= 70$$

$$\text{mode} = 90 - 3(90 - 80)$$

$$= 60$$

$$\text{skewness} = \frac{3(\text{mean} - \text{mode})}{\text{std deviation}}$$

$$= \frac{3(100 - 70)}{10}$$

$$= 9$$

$$\text{skewness} = \frac{3(\text{mean} - \text{mode})}{\text{std deviation}}$$

$$= \frac{3(90 - 60)}{10}$$

$$= 9$$

The given statement is true

3. The sum of 50 observations is 500 and sum of their squares is 6000 and the median is 12. Compute the coefficient of variation and coeff of skewness.

$$\sigma^2 = \frac{\sum x^2}{n} - \bar{x}^2$$

$$= \frac{6000}{50} - 100 = 20$$

$\bar{x} = 100$ (using data)

$$\sigma = \sqrt{20}$$

$$= 4.472$$

$$CV = 100 \times \frac{\sigma}{\bar{x}} = \frac{100 \times 4.47}{10}$$

INCOMPLETE

4. Compute the first 4 moments about mean from the following data, assumed mean = 25

CI	f	x	d = $\frac{x-A}{h}$	fd	(fd) ²	(fd) ³	fd	(fd) ²	(fd) ³	(fd) ⁴
0-10	1	5	-2	-2	4	-8	-2	4	-8	16
10-20	3	15	-1	-3	9	-27	-3	9	-27	81
20-30	4	25	0	0	0	0	0	0	0	0
30-40	2	35	1	2	4	8	2	4	8	16
$\Sigma f = 10$										

$$\mu'_1 = \frac{h \Sigma fd}{N}$$

$$\mu'_2 = \frac{h^2 \Sigma fd^2}{N}$$

$$\mu'_3 = \frac{h^3 \Sigma fd^3}{N}$$

$$\mu'_1 = 10 \times \frac{-3}{10} = -3$$

$$\mu'_2 = 100 \times \frac{17}{10} = 170$$

$$\mu'_3 = 1000 \times \frac{-27}{10} = -2700$$

$$\mu'_4 = \frac{10000 \times 113}{10} = 113000$$

Moments about mean

$$\mu'_1 = \frac{h \Sigma fd}{N}$$

$$\mu'_2 = \frac{h^2 \Sigma fd^2}{N}$$

$$\mu'_3 = \frac{h^3 \Sigma fd^3}{N}$$

if not given CI

$$\mu'_1 = \frac{\Sigma fd}{N}$$

$$\mu_1 = 0$$

$$\mu_2 = \mu'_2 - (\mu'_1)^2$$

$$\mu_3 = \mu'_3 - 3\mu'_2\mu'_1 + 2(\mu'_1)^3$$

$$\mu_4 = \mu'_4 - 4\mu'_3\mu'_1 + 6\mu'_2(\mu'_1)^2 - 3(\mu'_1)^4$$

$$\mu_1 = 0$$

$$\mu_2 = \mu'_2 - (\mu'_1)^2 = 170 - (-3)^2 = 161$$

$$\mu_3 = \mu'_3 - 3\mu'_2\mu'_1 + 2(\mu'_1)^3 = -2700 - 3(170)(-3) + 2(-3)^3 = -2700 + 1530 - 54 = -1224$$

$$\begin{aligned}\mu_4 &= \mu_4' - 4\mu_3'\mu_1' + 6\mu_2'(\mu_1')^2 - 3(\mu_1')^4 \\ &= 113000 - 4(-2700)(-3) + 6(170)(-3)^2 - 3(-3)^4 \\ &= 89537\end{aligned}$$

7/4/26 Moments

Statistic moments are arithmetic mean of second and third order.

They are statistical measure that give certain characteristics of data distribution. Moments are classified into raw moments and central moments.

The raw moments are measured about any arbitrary point A , if A is 0 then the raw moments are moments about origin. If A is arithmetic mean then we get central moments.

The first raw moment about origin is the mean. First central moment is zero.

Types of moments

- Moment about arbitrary point
- Moment about mean
- Moment about origin

Q1. Calculate the first 4 moments for the following data

X	1	2	3	4	5	6	7	8	9
f	1	6	13	25	30	22	9	5	2

assumed mean = 5

A = 5						
X	f	d = X - A	fd	(fd) ²	(fd) ³	(fd) ⁴
1	1	-4	-4	16	-64	256
2	6	-3	-18	324	-5832	104976
3	13	-2	-26	676	-17576	456976
4	25	-1	-25	625	-15625	390625
5	30	0	0	0	0	0
6	22	1	22	484	10648	234256
7	9	2	18	324	5832	104976
8	5	3	15	225	3375	50625
9	2	4	8	64	512	4096
113			$\sum fd = -10$	$\sum fd^2 = 2738$	$\sum fd^3 = -18730$	1346786

$$\mu'_1 = \frac{\sum fd}{N} = \frac{-10}{113} = -0.088$$

$$\mu'_4 = \frac{\sum (fd)^4}{N} = 11918.4$$

$$\mu'_2 = \frac{\sum (fd)^2}{N} = \frac{2738}{113} = 24.23$$

$$\mu'_3 = \frac{\sum (fd)^3}{N} = -165.7522$$

$$\mu_1 = 0$$

$$\mu_2 = \mu'_2 - (\mu'_1)^2 = (24.23) - (-0.088)^2 = 24.22$$

$$\begin{aligned} \mu_3 &= \mu'_3 - 3\mu'_2\mu'_1 + 2(\mu'_1)^3 = (-165.7522) - 3(24.23)(-0.088) \\ &\quad + 2(-0.088)^3 \\ &= -159.35 \end{aligned}$$

$$\begin{aligned} \mu_4 &= \mu'_4 - 4\mu'_3\mu'_1 + 6\mu'_2(\mu'_1)^2 - 3(\mu'_1)^4 \\ &= 11918.46 - 4(-165.75)(-0.088) - 3(-0.088)^4 \\ &\quad + 6(24.23)(-0.088)^2 \\ &= 11861.24164 \end{aligned}$$